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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis			
Date of Submission	2026-06-04		

Reviewer

Name	Dr. Yapeng Wang	Date	2026-06-19
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

The thesis presents a comprehensive deep learning-based framework for multi-cancer screening using chest CT scans, specifically focusing on pulmonary nodules and incidental thyroid cancer. It effectively addresses clinical bottlenecks such as longitudinal data scarcity and sequence feature redundancy through innovative architectures like the CL-IJLO framework and the DVT model. The research successfully transitions across spatial, temporal, and modal dimensions to optimize diagnostic workflows. Overall, the thesis is a well-structured and highly relevant contribution to the field of intelligent medical image analysis.

Knowledge Base and Literature Review.

The candidate demonstrates a robust understanding of medical imaging, tracing the paradigm shift from traditional radiomics to modern deep learning methodologies. The literature review systematically evaluates current artificial intelligence applications in thyroid and lung cancer diagnostics, accurately identifying gaps such as small sample sizes and modality

heterogeneity. By thoroughly discussing state-of-the-art techniques like Vision Transformers and contrastive learning, the review establishes a solid theoretical foundation that clearly justifies the clinical and technical motivations for the proposed multi-cancer screening framework.

Originality and Independence.

The research exhibits substantial originality, particularly in its methodological innovations for handling complex clinical imaging constraints. The introduction of a secondary transfer learning strategy combined with a Transformer architecture uniquely tackles the issue of high-noise and small-sample data in incidental thyroid cancer screening. Furthermore, the contrastive learning-based feature compression module and the combined contrastive loss function represent highly independent, novel approaches to purifying longitudinal CT sequence redundancy and aligning model optimization with clinical recall requirements.

Arguments and Reasoning.

The candidate's arguments are logically sound and heavily supported by empirical evidence. The thesis meticulously details experimental designs, including multi-center data collection and rigorous ablation studies, to validate the necessity of each proposed module. For instance, the ablation experiments on the CL-IJLO model clearly demonstrate the synergistic effect of the compression module and joint loss function in balancing diagnostic sensitivity and specificity. The reasoning is progressive, effectively bridging technical artificial intelligence metrics with real-world clinical utility and decision-making workflows.

Evaluation within the Field.

Within the field of medical image analysis, this work represents a significant step toward practical, multi-target intelligent screening. The DVT model achieved an impressive AUC of 0.97 on enhanced CT datasets, proving the viability of opportunistic thyroid screening. Similarly, the CL-IJLO framework demonstrated excellent predictive capability for pulmonary nodule malignancy, outperforming several state-of-the-art baseline models like SMILE and GSC-DVITD in sensitivity and overall robustness. The exploratory work on multimodal fusion also provides valuable baseline metrics for future precision medicine architectures.

Completeness and Coherence.

The thesis is highly coherent and structurally complete, seamlessly linking the spatial, modal, and temporal dimensions of chest CT analysis into a unified framework. Each chapter builds logically upon the previous ones, starting from a broad literature review, transitioning into specific spatial thyroid screening and temporal lung nodule prediction models, and culminating in an exploratory multimodal synthesis. The dual-stage algorithm designs, such as the adjacent image splicing in the DVT model, show a complete and thoughtful approach to optimizing both computational efficiency and diagnostic accuracy.

Conclusion.

In conclusion, this thesis presents a highly commendable exploration of deep learning frameworks for chest CT-based multi-cancer screening. The candidate successfully formulated algorithms that address critical real-world issues like sequence redundancy and small-sample noise, verifying them through comprehensive multi-center datasets. While honestly acknowledging limitations like dataset diversity and multimodal fusion bottlenecks, the proposed future roadmap is academically rigorous. This work makes substantial contributions to medical artificial intelligence, offering reliable, data-driven auxiliary tools for early cancer diagnosis and clinical resource optimization.

Additional Comments.